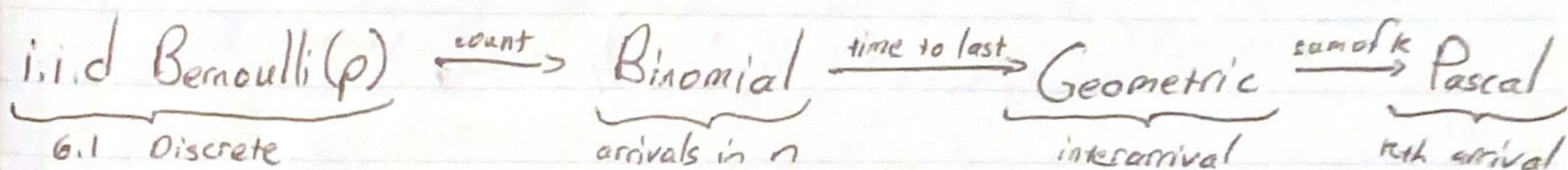


RES.6-012 Intro To Probability - Lectures 21, 22, 23
Textbook Notes - Sections 6.1 & 6.2

• Section 6.1 The Bernoulli Process

• Architecture



• Definition

Independent $X_1, X_2, \dots$ with $P(X_i = 1) = p$

• Associated RVs

Successes S in n trials, Binomial(n, p):

$$p_S(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad E[S] = np, \quad \text{var}(S) = np(1-p)$$

Time T to first success, Geometric(p):

$$p_T(t) = (1-p)^{t-1} p, \quad E[T] = \frac{1}{p}, \quad \text{var}(T) = \frac{1-p}{p^2}$$

• Independence & Memorylessness

- Disjoint-trial Rule: RVs defined on non-overlapping trial sets are independent.
- Fresh start: $X_{n+1}, X_{n+2}, \dots$ is a Bernoulli process independent of $X_1, \dots, X_n$
- Memorylessness: For T the first success after n :
$$P(T-n = t \mid T > n) = (1-p)^{t-1} p = P(T=t)$$

↳ Fresh Start at random time $N \rightarrow$ Holds if N depends only on past. ~~FF~~

↳ Method: condition on $\{N=n\}$

• Interarrival Times & kth Arrival

Interarrival times $T_1, T_2, \dots$ i.i.d. Geometric(p) (equivalent)

$$Y_k = T_1 + \dots + T_k \rightarrow \text{Pascal order } k$$

$$P_{Y_k}(t) = \binom{t-1}{k-1} p^k (1-p)^{t-k}, \quad t \geq k$$

$$E[Y_k] = \frac{k}{p} \quad \text{var}(Y_k) = \frac{k(1-p)}{p^2}$$

Method: $\{Y_k = t\} = \text{success at } t \text{ AND } k-1 \text{ successes in } t-1;$
independent (so multiply).

• Splitting & Merging

→ Split: (keep w.p. q): kept arrivals Bernoulli(pq),
discarded Bernoulli($p(1-q)$)

→ Merge: ~~two~~ two independent processes p, q : Bernoulli($p+q-pq$)

• Poisson Approximation to Binomial

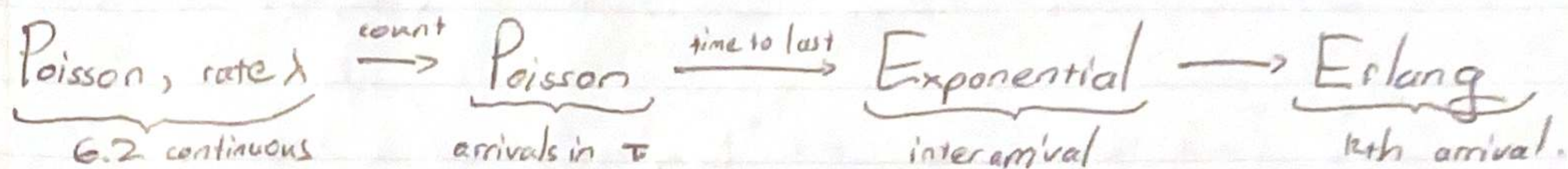
As $n \rightarrow \infty$, $p \rightarrow 0$, $\lambda = np$ fixed.

$$\underbrace{\binom{n}{k} p^k (1-p)^{n-k}} \rightarrow \underbrace{e^{-\lambda} \cdot \frac{\lambda^k}{k!}}$$

↳ Good for $\{n \geq 100, p \leq 0.01\}$

• Section 6.2 The Poisson Process

• Architecture



↳ Poisson is continuous-time limit of Bernoulli.

↳ Every 6.2 result mirrors a 6.1 result under $\{p \rightarrow \lambda \delta, \delta \rightarrow 0\}$

• Definition

$P(k, \tau)$ = probability of k arrivals of length τ .
Poisson with rate λ if:

- (a) Time Homogeneity: $P(k, \tau)$ same for all intervals τ of length τ .
- (b) Independence: Disjoint intervals independent.
- (c) Small interval probabilities

$$P(0, \tau) = 1 - \lambda\tau + o(\tau), \quad P(1, \tau) = \lambda\tau + o(\tau), \\ P(k, \tau) = o_k(\tau), \quad k \geq 2$$

Derivation: partition $[0, \tau]$ into $n = \frac{\tau}{\delta}$ slots with $p = \lambda\delta$; let $\delta \rightarrow 0$ with $np = \lambda\tau$ fixed.

• Associated RVs

Arrivals N_r in length $\tau \rightarrow \text{Poisson}(\lambda\tau) :$

$$P(k, \tau) = e^{-\lambda\tau} \cdot \frac{(\lambda\tau)^k}{k!}, \quad E[N_r] = \text{var}(N_r) = \lambda\tau$$

Time T to first arrival $\rightarrow \text{Exponential}(\lambda) :$

$$f_T(t) = \lambda e^{-\lambda t}, \quad E[T] = \frac{1}{\lambda}, \quad \text{var}(T) = \frac{1}{\lambda^2}$$

From $F_T(t) = 1 - P(0, t)$

$P(k, T)$ applies to any set of times totalling T , not just one interval.

• Independence & Memorylessness

$\rightarrow$ Fresh start : Process after t is Poisson, independent of history to t .

$\rightarrow$ Memorylessness : For T the first arrival after t :
 $T - t \sim \text{Exponential}(\lambda)$,
 since $P(T - t > s) = P(0, s) = e^{-\lambda s}$

• Interarrival Times & k th Arrival

$T_1, T_2, \dots$ i.i.d. $\text{Exponential}(\lambda)$ (equivalent)

$Y_k = T_1 + \dots + T_k \rightarrow \text{Erlang order } k :$

$$f_{Y_k}(y) = \frac{\lambda^k y^{k-1} e^{-\lambda y}}{(k-1)!} \quad ; \quad y \geq 0$$

$$E[Y_k] = \frac{k}{\lambda} \quad ; \quad \text{var}(Y_k) = \frac{k}{\lambda^2}$$

Method : arrival in $[y, y+\delta]$ (prob $\approx \lambda\delta$) AND $k-1$ arrivals before y (prob. $P(k-1, y)$) ; multiply, divide by δ

Exact Alternate : $F_{Y_k}(y) = 1 - \sum_{n=0}^{k-1} P(n, y)$, then differentiate

• Splitting & Merging

→ Split : (keep w.p. p) kept Poisson(λp)
discarded Poisson($\lambda(1-p)$)
↳ mutually independent

→ Merge : rates $\lambda_1, \lambda_2 \rightarrow \text{Poisson}(\lambda_1 + \lambda_2)$.

→ Any arrival can come from the first with prob.
 $\frac{\lambda_1}{\lambda_1 + \lambda_2}$

independently across arrivals
→ Extends to n processes

Method for both : verify the three defining properties using small-interval probabilities.

• Competing Exponentials

$$\min\{T_a, T_b\} \sim \text{Exponential}(\lambda_a + \lambda_b)$$

Method : first arrival of the merged process.

Last of n : For n i.i.d. Exponential(λ) :

• total time = $\sum_{j=1}^n T_j$ with $T_j \sim \text{Exponential}(j\lambda)$,
independent by memorylessness.

• So $E[\text{total}] = \sum_{j=1}^n \frac{1}{j\lambda}$

• Sums of a Random Number of RVs

$$Y = X_1 + \dots + X_N, \quad \text{all independent:}$$

<u>X_i</u>	<u>N</u>	<u>Y</u>
Bernoulli(p)	Binomial(m, p)	Binomial(m, pq)
Bernoulli(p)	Poisson(λ)	Poisson(λp)
Geometric(p)	Geometric(q)	Geometric(pq)
Exponential(λ)	Geometric(q)	Exponential(λq)

↳ All follow from splitting. (Transform derivations in Sec 4.4)

• The Random Incidence Paradox

For fixed t^* , the interarrival interval L containing it is Erlang order 2, not exponential:

$$E[L] = \frac{2}{\lambda}$$

Why? $L = (t^* - U) + (V - t^*)$ (elapsed time + remain time)

- ↳ Independent (past vs. future)
- ↳ each Exponential(λ) (remaining by memorylessness)
- ↳ elapsed by time-reversal symmetry
- ⇒ Sum is Erlang order 2.

An arbitrary instant is more likely to land in a long ^{interval} ~~time~~.
 "Typical interval" is undefined until the selection mechanism is specified.

• Bernoulli $\leftrightarrow$ Poisson Correspondence

	Bernoulli	Poisson
Time	discrete slots	continuous
Rate	p per trial	λ per time unit
Arrivals in interval	Binomial	Poisson
Interarrival time	Geometric	Exponential
Time to k th arrival	Pascal order k	Erlang order k
Split w.p. q	Bernoulli(pq)	Poisson(λq)
Merge	$p + q - pq$	$\lambda_1 + \lambda_2$